

SAMUEL MAXWELL OBENG AVORNYOH

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PROFESSIONAL SUMMARY

- AI/ML Engineer with over four years of experience developing Machine Learning models, AI systems, data-driven platforms, automation tools, and machine learning solutions for practical business and technical problems.
- Strong experience building reliable data pipelines, AI agents, forecasting models, full-stack web apps, embedded-system design tools, and business management systems.
- Excellent technical communication skills with experience documenting system architecture, explaining complex workflows, and collaborating in remote or cross-functional environments.

SKILLS

Technical Skills: Data Engineering, AI Systems, MLOps, Blockchain, Machine Learning, Embedded System Design, Networking, Game Development, Data Visualization

Programming Languages: Python, Rust, JavaScript/TypeScript, SQL, Bash, Solidity, Go, C/C++

AI/ML Frameworks: PyTorch, scikit-learn, LangChain, CrewAI, MLflow, RAG, Vector Databases

Tools & Technologies: Google Cloud Platform, AWS, Docker, Kubernetes, Git/GitHub, LLMs, ANNOY, Chroma, Prometheus/Grafana

Backend Services & Databases: FastAPI, Express.js, Axum/Rust, Redis, PostgreSQL/Supabase, Neo4j, RabbitMQ, Firebase

Soft Skills: Adaptability, Problem-solving, Effective Communication, Collaboration, Leadership, Technical Documentation

EDUCATION

BSc. Computer Science and Engineering | University of Mines and Technology, Tarkwa Ghana

Academic Standing: Second Upper | CWA: 75.66

Relevant Coursework: Artificial Intelligence, Data Structures and Algorithms, Database Systems, Data Science Fundamentals, Probability and Statistics, Embedded System Design, Linear Algebra, Robotics

PROFESSIONAL EXPERIENCE

TonyCold Store Management System | Full-Stack Developer

May 2026 - Present

- Developed a business management system for a cold store dealing in frozen meat, fish, and related products.
- Designed modules for product records, inventory tracking, sales/POS operations, stock movement, expense tracking, and business reporting.
- Improved operational visibility by organizing product categories, pricing, sales summaries, and admin workflows for day-to-day store management.

Machine Learning Intern | Makersplace

Sep. 2023 – Jan. 2024

- Created a web-based generative AI assistant to support robotic facilitators in course delivery using a facilitator guidebook as a knowledge source for contextual responses.
- Built visual charts and graphs to communicate AI-agent activities, performance, and decision insights to non-technical stakeholders.
- Developed machine-learning models for financial-market forecasting and trading-strategy research using predictive algorithms and market indicators.
- Integrated AI agents with Telegram and WhatsApp using FastAPI, Telegram libraries, and WhatsApp automation tools.

Blockchain / Gaming / AI Project Developer | Independent Projects

May 2025 – Present

- Built game and blockchain prototypes including arcade-style games, Web3/crypto experiments, and AI-assisted development workflows.
- Facilitated Web3, forex, and AI learning discussions across WhatsApp and LinkedIn communities.
- Continued building portfolio projects in AI agents, embedded systems, trading systems, and full-stack web platforms.

National Service Personnel / AI & Data Engineer | Really Great Tech

Dec. 2025 – Present

- Built and maintained an AI/Data training repository covering data analytics, SQL, dashboards, supervised ML, deployment, MLOps, LLM fundamentals, LangChain, RAG, and model evaluation.
- Designed weekly workflows and milestone projects using Git/GitHub branches, pull requests, Python, Jupyter/VS Code, scikit-learn, SQL, and Google Looker Studio.
- Developed learning materials and project documentation for business insights, ML microservices, and telecom policy assistant use cases.

PERSONAL PROJECTS

AI / ML, Blockchain, networking & Full-Stack Project Portfolio

Nov. 2022 – Present

- Developed **FSM Designer / NeuroBench**, a graphical finite-state-machine editor for embedded systems with simulation, templates, C/Python export, and AI-assisted debugging.
- Explored a **Chain Registry Ecosystem** project focused on registry integrity, blockchain metadata management, reproducible verification, governance, auditability, and supply-chain security.
- Designed **proxy marketplace** and recommendation-feed algorithm prototypes involving networking, marketplace architecture, usage-based billing, retrieval, ranking, scoring, and personalization.
- Designed a **Crypto POS payment system** concept for merchants to accept crypto payments, support stablecoins and major assets, verify transactions, generate receipts, and manage fiat or crypto settlement workflows.
- Built a **healthcare no-show prediction** system using FastAPI, React, MLflow, Redis, RabbitMQ, PostgreSQL, and Prometheus/Grafana monitoring.
- Built **AI Merchant Assistant**, a Rust/Next.js/Supabase platform for voice-based sales recording, receipt OCR, inventory tracking, real-time analytics, forecasting, and smart merchant alerts.
- Developed **sports odds arbitrage** analytics tools with Python scraping, Redis caching, fuzzy/NLP event matching, alerting, rate limiting, and reliability-focused fallback handling.
- Developed an AI-powered music companion platform with React/TypeScript, Express.js, Supabase, Gemini, Spotify, YouTube/yt-dlp, Telegram Bot integration, and real-time WebSocket features.
- Designed a hybrid CNN-LSTM sign-language detection pipeline using video/image data and pose-based features.

ACHIEVEMENTS

- One of the 50 winners of Google DeepMind - Vibe Code with Gemini 3 Pro in AI Studio Hackathon. Link: [Hackathon Winners](#). [Hackathon project](#)